

CoinQuest

A gamified expense tracker with a local AI coach — log, level up, and conquer the month

ONE-PAGE SOFTWARE SPEC v1.0 · 12 JUL 2026
AI Engineering Summer Training 2026
Week 2 — AI-assisted development · Day 1

Budget HP

1500 / 1500 SAR

Streak
Day 0

XP
0

01 THE PROBLEM IT SOLVES

Expense trackers have a retention problem: logging what you spend is a chore with no feedback loop, so most people quit within a week — and students rarely find out where the allowance went. **CoinQuest reframes the habit as a game:** the monthly budget is a health bar, consistent logging builds streaks and XP, and a local AI coach turns the numbers into advice — or a roast. Tracking money should feel like maintaining a save file, not filing taxes.

02 KEY FEATURES

COMMON · CORE SLICE	built day 2 — one thin path through every layer
Log an expense	amount, category, note, date → POST /api/expenses → SQLite
Expense list	newest first from GET /api/expenses — proof the full stack is wired
RARE · V1.0	shipped by day 5
Budget HP bar	drains in real time; green → yellow → red, and 0 HP triggers a "defeated" state
Streak & XP	consecutive logging days keep the streak alive; every entry earns XP
Monthly summary	backend computes totals at GET /api/summary ; Chart.js draws the charts
AI coach	Ollama turns pre-computed stats into a review + one tip — Coach / Roast / Pirate modes
Monthly boss fight	top spending category becomes the month's boss — "Restaurant Monster dealt 720 SAR"
EPIC · STRETCH	only if day 5 has hours left
If time allows	AI expense entry ("Lunch at AlBaik 28 SAR" → structured JSON) · AI weekly quests · achievements & titles — full detail in SPEC.md

TECH LOADOUT

FastAPI · Python 3

SQLite

HTML · CSS · JS

Chart.js

Ollama local LLM

Git + GitHub · tags v0.1 → v1.0

03 CONSTRAINTS — WHAT IT WILL NOT DO

- ✗ **No accounts, logins, or multi-user.** One player, on localhost.
- ✗ **No bank sync, card data, or real financial integrations.**
- ✗ **No cloud deployment.** Data never leaves the machine.
- ✗ **The AI never does arithmetic.** Python computes every number; the model only writes text.
- ✗ **AI output is advisory only.** It never creates, edits, or deletes records.
- ✗ **One person + one AI assistant, five days.** Features that threaten the deadline move down a tier.

DEFINITION OF DONE

- ☐ Clone, run one command, add an expense, watch the HP bar drop.
- ☐ Monthly summary shows real charted totals plus the AI coach's boss report.
- ☐ Repo history tells the week's story: daily tags v0.1 → v1.0, REVIEW.md , BUGLOG.md .